POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

FACULTY OF ARTIFICIAL INTELLIGENCE

Deep Learning – EndTerm Report

Project Name: GraphCast SEA

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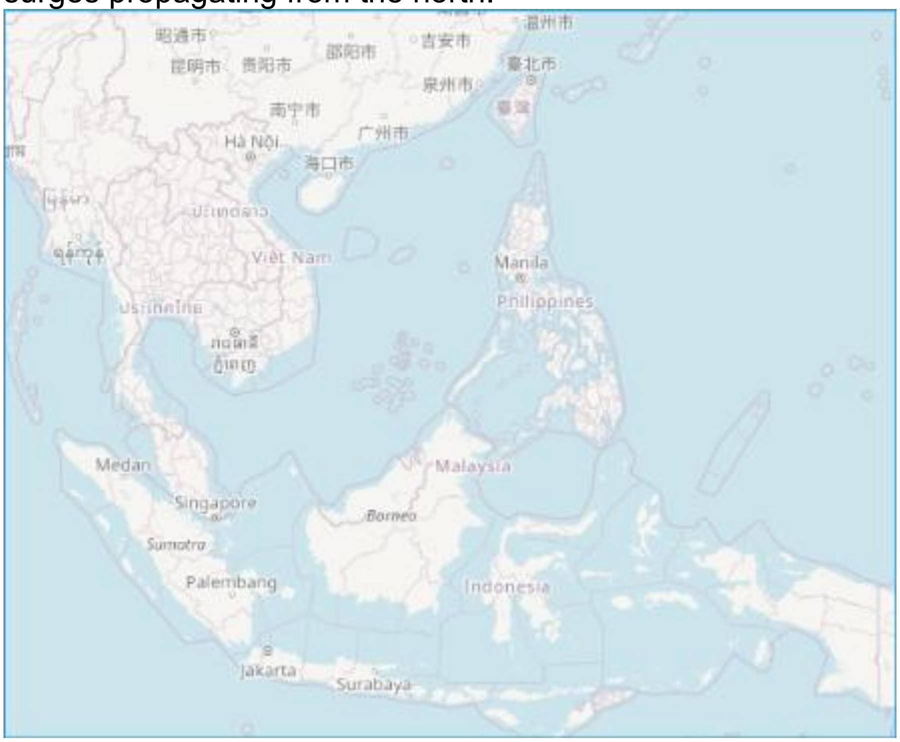
Contents

1. Overview

1.1 Spatial Scope

The study region is confined to Southeast Asia (latitudes −11° to 28,5°, longitudes 92° to 142°), encompassing Vietnam, Laos, Cambodia, Thailand, Malaysia, Indonesia, the Philippines, southern China, a portion of the Indian Ocean, and the Western Pacific. This coverage is sufficiently broad to encapsulate the weather systems that directly impact Vietnam, such as tropical cyclones originating from the Western Pacific or cold surges propagating from the north.



At a horizontal resolution of $\Delta = 0.25^\circ$, the regular latitude - longitude grid contains

$$N_{\text{lat}} = \frac{28.5^\circ - (-11^\circ)}{0.25^\circ} + 1 = 159, \qquad N_{\text{lon}} = \frac{142^\circ - 92^\circ}{0.25^\circ} + 1 = 201$$

for a total of N = Nlat × Nlon = 31,959 grid nodes. This node count is the single most important quantity governing the computational design: it is precisely because N ∼ 3 × 104 that the O(N2) cost of dense attention becomes prohibitive and a graph-based, sparse-connectivity formulation becomes attractive

1.2 Key Parameters

- Number of forecast variables: 14 dynamic variables
- Spatial resolution: 0,25° × 0,25° (~28 km)
- Temporal frequency: Every 6 hours (at 00, 06, 12, and 18 UTC)
- **Dual input data sources:** ERA5 reanalysis (Brain branch) and NASA GPM

MERGIR satellite imagery (Eye branch)

2. Data Sources

2.1 ERA5 — The Brain Branch

ERA5 is the fifth-generation atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). It is constructed by combining a numerical weather prediction (NWP) model with data assimilation from all global observational sources, including satellites, radiosondes, aircraft, and surface stations. The result is a gridded, dynamically consistent, and gap-free dataset across all variables, grid points, and timesteps—widely recognized within the meteorological community as the closest approximation to 'ground truth' currently available.
Rationale for Selecting ERA5 Data: ERA5 is the gold-standard dataset for state-of-the-art meteorological machine learning research, utilized by leading models such as GraphCast, Pangu-Weather, and FourCastNet. It is accessible via the Copernicus Climate Data Store (CDS) API, provides a comprehensive historical record dating back to 1940, and offers a standardized, regular grid format that is highly conducive to model ingestion and processing.

ERA5 Variables
Surface variables:

- t2m — 2-meter air temperature (in Kelvin)
- d2m — 2-meter dewpoint temperature, combined with t2m to calculate relative humidity
- u10**,** v10 — eastward and northward components of wind at 10 meters (or *u- and v-components of 10-meter wind*)
- msl — mean sea-level pressure, critical for identifying low-pressure systems, cyclones (storms), and fronts

Accumulated Variable:

- tp — total precipitation accumulated over the previous 6 hours (in meters)

Pressure-level Variables (measured at various atmospheric layers):

- z — geopotential, dividing by g (9.80665 m/s^2) yields the geopotential height
- q — specific humidity
- u**,** v — wind components (eastward and northward) at the 500 hPa and 850 hPa pressure levels

Rationale for Selecting 500 hPa and 850 hPa Pressure Levels: These two levels are standard meteorological reference points. The 850 hPa level (approximately 1,500 m) sits just above the planetary boundary layer, making it ideal for monitoring monsoons and moisture transport. The 500 hPa level (approximately 5,500 m) represents the mid-troposphere and is critical for governing the steering flow and movement of large-scale weather systems.

2.2 NASA GPM MERGIR — The Eye Branch

The MERGIR (Merged Geostationary Infrared) product blends imagery from multiple geostationary satellites (including Japan's Himawari-8, the US's GOES, and Europe's Meteosat), delivering a single variable: brightness temperature (Tb) at the 11 μm infrared channel, with a spatial resolution of 4 km and a temporal frequency of 30 minutes.

Data Interpretation Principle: In clear-sky regions, the satellite observes the warm surface of the Earth, resulting in high Tb values. Conversely, in cloudy areas, the satellite detects the cold cloud tops; higher cloud tops exhibit colder temperatures, leading to lower Tb values. This serves as a direct proxy for convective intensity.

Rationale for Integration Alongside ERA5: The total precipitation (tp) variable in ERA5 is a model-generated output prone to systematic biases in tropical regions. In contrast, Tb represents real-time, independent observational data from satellites, offering a significantly higher spatial resolution (4 km compared to 28 km). This 'Eye' branch enhances the model's capacity to recognize deep convection and tropical cyclones more effectively."

2.3 Dataset Splitting

The dataset spanning from 2020 to 2026 is partitioned chronologically to preserve the temporal continuity of atmospheric states and prevent data leakage:

Dataset	Time Period	Samples
Training (Train)	2020, 2021, 2022, 2023, 2024	7303
Validation (Val)	2025	1459
Testing (Test)	2026	587

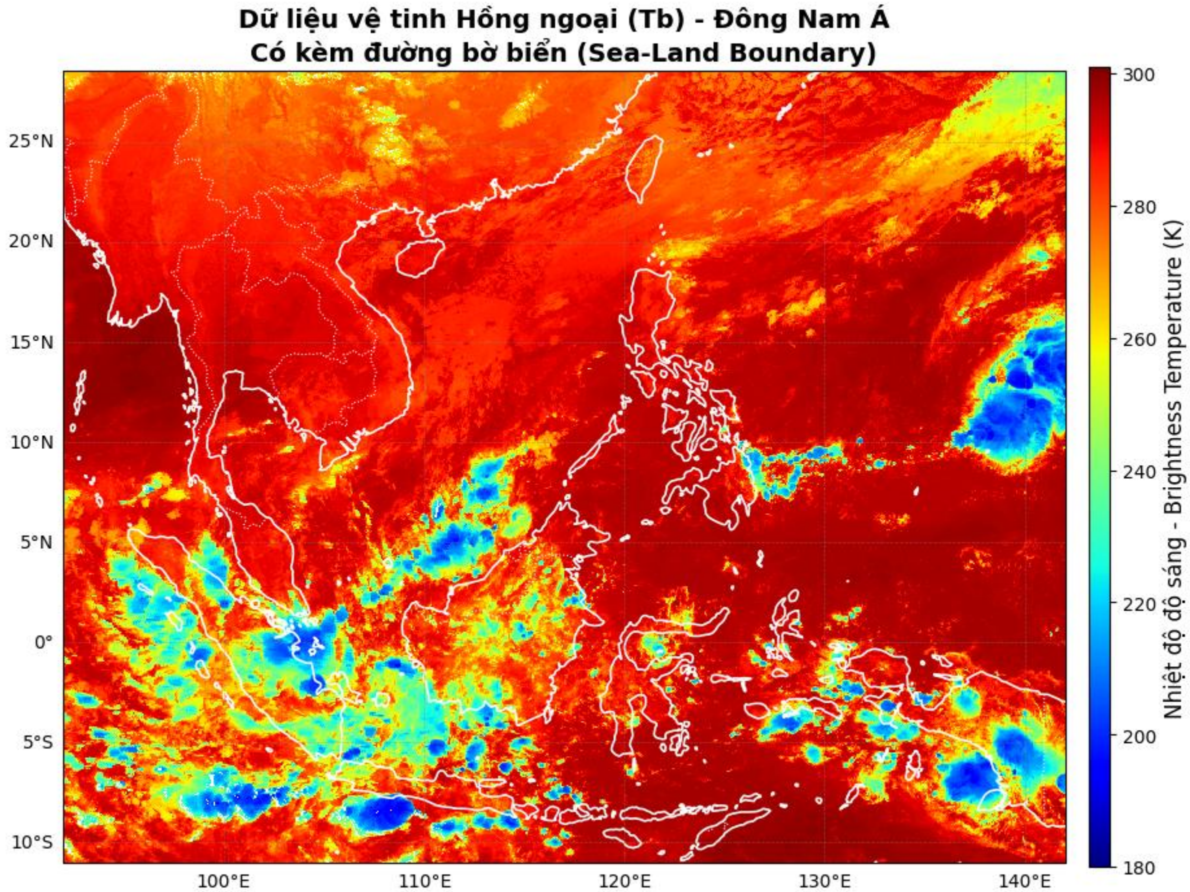
3. Pipeline

Module 1 — ERA5 Data Retrieval

Retrieve surface data (instantaneous and accumulated variables) and pressure level data via the CDS API. The data is subsequently cropped to the Southeast Asia region and exported in NetCDF format. Due to CDS quota limits, the pressure level data retrieval is split into two half-year periods.

Module 2 — NASA GPM Data Retrieval

Download .nc4 files from NASA GES DISC. Filter the time steps to 6-hour intervals (retaining only 00, 06, 12, and 18 UTC), crop to the Southeast Asia region, normalize longitudes, and export to Zarr format.
Example:



Module 3 — Data Preprocessing

- **1. ERA5 Harmonization:**
- **Lazy Loading:** Data is ingested in optimized chunks (time: 100) across three streams: 3D pressure levels, 2D surface accumulation, and 2D surface instantaneous variables.
- **Alignment:** The surface geopotential variable (z) is explicitly renamed to geopotential_surf to prevent naming collisions; system-generated auxiliary coordinates (e.g., expver) are removed.
- **Merging:** The streams are fused into a unified dataset (ds_brain) using an inner join on the temporal dimension.

- **2. Normalization & Label Generation:**
- **Z-Score Normalization:** Features are standardized using global statistics (Mean/Std) pre-computed solely from the 2020–2024 training baseline.

$$\tilde{X}^{(c)} = \frac{X^{(c)} - \mu_c}{\sigma_c}$$

- **Label Construction:** Future target labels are generated by shifting the normalized data backward by one timestep (shift(time=-1)), establishing a +6-

hour forecasting objective.

- **Serialization:** Processed data is exported as optimized Zarr stores: brain_{year}_x.zarr (inputs) and brain_{year}_label.zarr (targets).
- **Computing**: (μc, σc) on the training years only is a subtle but essential anti-leakage measure: using global statistics that include 2025 - 2026 would smuggle future information into the normalization constants. The same discipline that governs the data split must govern every fiitted preprocessing parameter.

- **3. NASA Satellite Processing:**
- **Temporal Alignment:** Raw satellite brightness temperature (Tb) imagery is filtered to match ERA5’s exact 6-hourly intervals (00, 06, 12, 18 UTC).
- **Spatial Resampling:** A linear interpolation algorithm maps the high-resolution satellite grid onto the 0.25° atmospheric grid of the ERA5 domain.
- **Standardization:** NASA data is standardized using global satellite distribution metrics and exported as eye_{year}_norm.zarr.

Module 4 — Model Training

- **Graph Construction:** Dynamically builds a hierarchical topology (Grid→Mesh, Mesh→Mesh, Mesh→Grid) to enable multi-scale spatial message passing.
- **Residual Learning:** Trains the model to predict temporal residuals ($\Delta X = X(t+1) - X(t)$ using a 2-timestep input history $X(t-1)$, $X(t)$ for improved temporal stability.
- **Weighted Optimization:** Employs a latitude-weighted RMSE loss function to mitigate the bias caused by grid convergence at higher latitudes.
- **Training & Checkpointing:** Utilizes early stopping based on validation loss and serializes the optimal model state as graphcast_sea_best.pt.

Module 5 — Evaluation and Visualization

The evaluation workflow assesses the predictive performance and generalization capabilities of the GraphCast SEA model through the following stages:

- **Autoregressive Rollout:** * Executes a long-range forecasting rollout starting from 30 distinct temporal points.
    - Each iteration proceeds for 12 autoregressive steps, allowing for the analysis of error accumulation over time.

- **Performance Metrics:** * Calculates RMSE for each atmospheric variable across all lead times.
    - Performs a comparative analysis against the **Persistence Baseline** (assuming future states equal current states) to quantify the model's actual forecasting gain.

- **Denormalization & Visualization:** * Reverts normalized predictions into physical units (K, Pa, m/s) using training statistics.

    - Visualizes training convergence via loss curves, validates RMSE trends, and generates spatial maps of both forecasted states and residual error distributions.

4. Model Structure

4.1. The Case for GNNs in Atmospheric Modeling

Standard CNNs are unsuitable for global atmospheric forecasting due to their assumption of translation invariance, which fails to account for the physical differences between equatorial and polar regions, or land and ocean surfaces. Furthermore, CNNs are constrained by fixed receptive fields and face significant challenges when applied to non-Euclidean spherical grids. Pure Transformer architectures are similarly impractical; full self-attention incurs O(n^2) computational complexity, which is prohibitively expensive given the tens of thousands of grid points in high-resolution atmospheric data.

Graph Neural Networks (GNNs) are the superior choice, as they do not assume spatial homogeneity, allow for flexible long-range edge connections, and efficiently model irregular, multi-resolution spatial structures with linear scaling relative to the number of nodes.

4.2 Grid-Mesh-Grid Structure (Encoder–Processor–Decoder).

The GraphCast SEA model utilizes a hierarchical multi-scale topology structured into three primary components:

- **Encoder (Grid→Mesh):** This stage projects observational data from the regular 0,25° latitude-longitude grid onto a learnable, sparse icosahedral mesh. Each grid node is connected to its kNN=4 nearest mesh nodes. A Multilayer Perceptron (MLP) encodes the input atmospheric state into the latent mesh space:

$$\mathbf{h}_i^0 = \mathrm{MLP}_{\mathrm{node}}(\mathbf{x}_i), \qquad \mathbf{e}_{ij}^0 = \mathrm{MLP}_{\mathrm{edge}}(\mathbf{e}_{ij})$$

$$\mathbf{m}_{ij} = \mathrm{MLP}([\,\mathbf{h}_i^0 \,\|\, \mathbf{h}_j^0 \,\|\, \mathbf{e}_{ij}^0\,]), \qquad \mathbf{h}_j^{\mathrm{mesh}} = \mathrm{MLP}\Big(\big[\,\mathbf{h}_j^0 \,\|\, \sum_{i \in \mathcal{N}(j)} \mathbf{m}_{ij}\,\big]\Big)$$

- **Processor (Mesh→Mesh):** The core of the model consists of a stack of 4 GNN layers that perform an iterative message passing over the icosahedral mesh. By exchanging information between neighboring mesh nodes, the processor effectively captures complex, non-linear atmospheric dynamics across multi-scale spatial dependencies:

$$\mathbf{e}_{uv}^{\ell+1} = \mathbf{e}_{uv}^{\ell} + \phi_e\big([\,\mathbf{h}_u^{\ell} \,\|\, \mathbf{h}_v^{\ell} \,\|\, \mathbf{e}_{uv}^{\ell}\,]\big),$$
$$\mathbf{h}_v^{\ell+1} = \mathbf{h}_v^{\ell} + \phi_h\Big([\,\mathbf{h}_v^{\ell} \,\|\, \sum_{u\in\mathcal{N}(v)} \mathbf{e}_{uv}^{\ell+1}\,]\Big).$$

- **Decoder (Mesh→Grid):** This component maps the latent representations from the mesh back to the physical observational grid. In this residual-learning configuration, the model outputs the predicted atmospheric residual ($\Delta X = X(t+1) - X(t) which is then added to the current state X(t) to generate the final X(t+1) forecast:

$$\Delta\widehat{X} = g_\theta\big(\mathbf{h}^{\text{mesh}}\big), \qquad \widehat{X}_{t+1} = X_t + \Delta\widehat{X}$$

5. Training Configuration

5.1 Model Configuration

- Hidden dimension (hidden dim): 128
- Number of processor layers: 4
- Activation function: SiLU
- Normalization: LayerNorm
- Number of parameters: 538,382 parameters

5.2 Training Configuration

- Optimizer: AdamW (weight_decay=1e-5)
- Learning rate: 1e-4, CosineAnnealingLR decaying continuously across 50 epochs
- Batch size: 4
- Epochs: 50
- Early stopping: patience=7 epoch on val loss
- Loss: Latitude-weighted RMSE

$$wRMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} w_i \cdot (y_i - \hat{y}_i)^2}$$

$$w_i = \frac{\cos(\phi_i)}{\frac{1}{M}\sum_{j=1}^{M}\cos(\phi_j)}$$

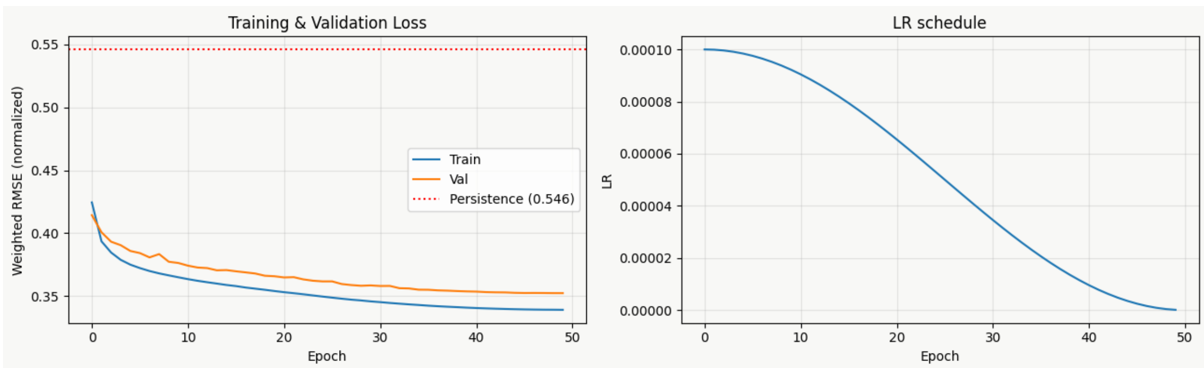
5.3 Rationale for Utilizing Latitude-Weighted RMSE

Although a regular 0,25° latitude-longitude grid is evenly spaced in terms of degrees, it is highly non-uniform with respect to actual physical surface area. For instance, 1

longitude spans approximately 111 km at the equator, but shrinks to only about 55 km at 60° latitude.
Without appropriate weighting, grid points at higher latitudes would be over-represented in the loss function relative to their actual spatial coverage. Introducing a cos(lat) weighting factor effectively counterbalances this geometric distortion, ensuring that the model optimizes uniformly across the entire target domain.

6. Results

6.1 Training Result



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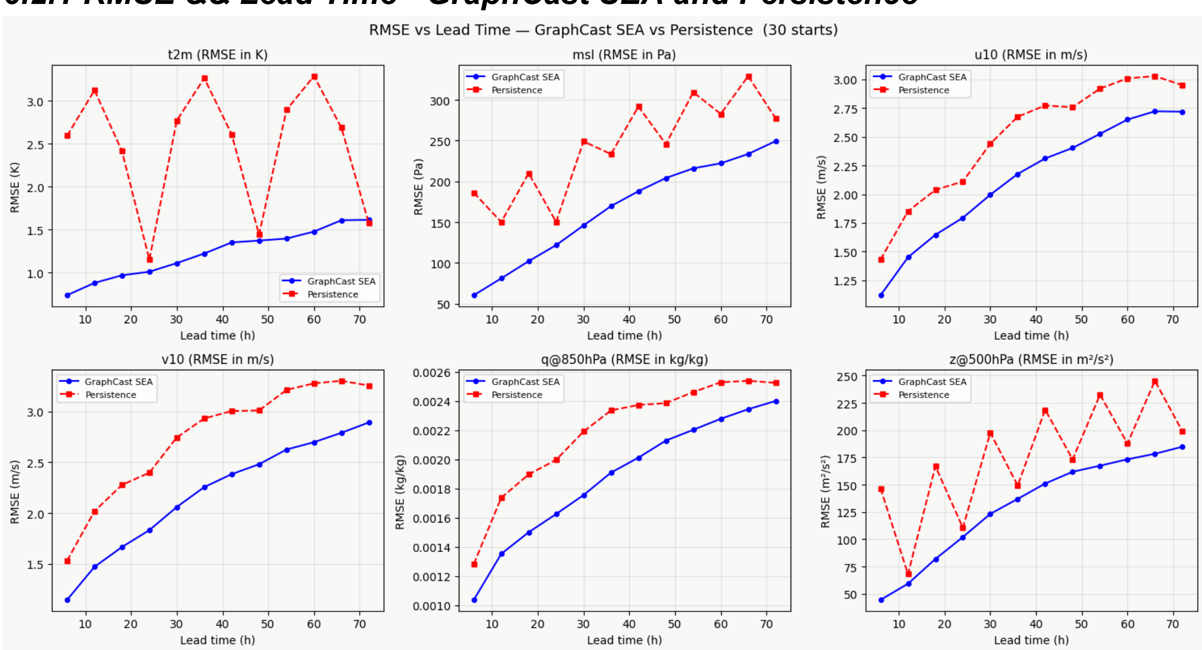
Following the diagram above, we can clearly see that GraphCast model outperforming persistent baseline in both training and validating process -> result in a much better outcome for the project

6.2 Evaluation Results

6.2.1 RMSE && Lead Time - GraphCast SEA and Persistence



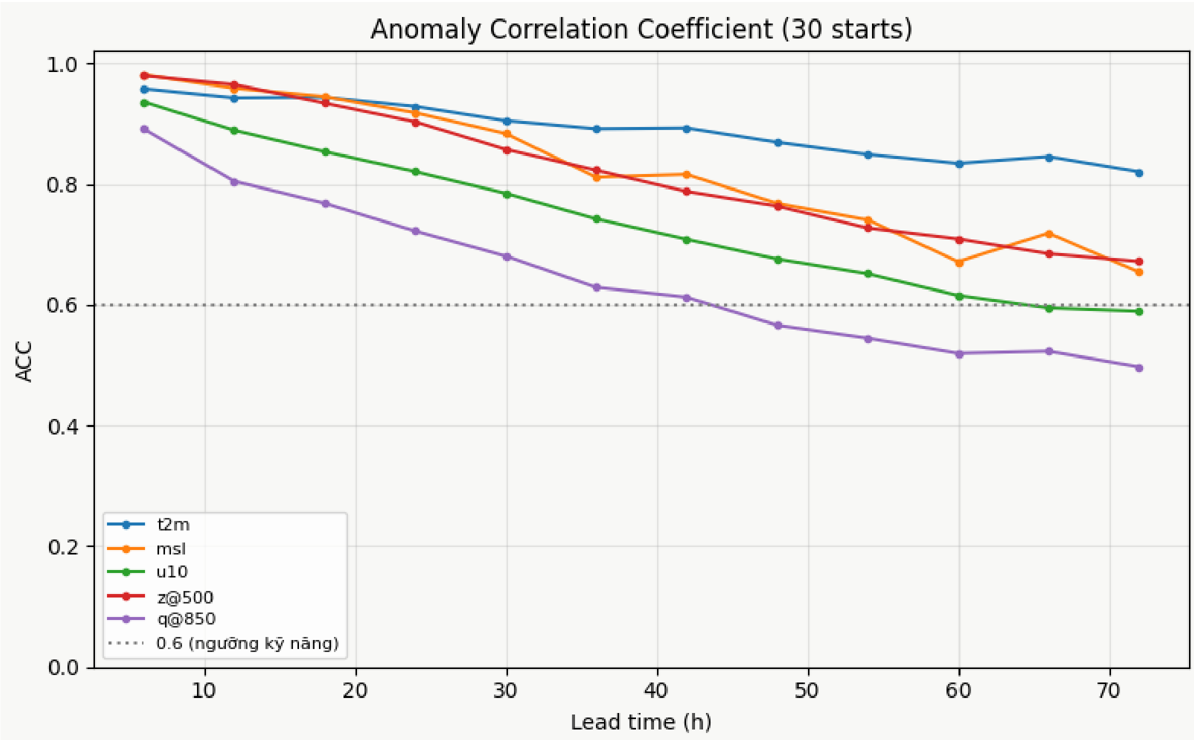
Group 1: Dominant Model Variables (Unimpacted by Diurnal Cycles)
Core Characteristic: Error curves for both the model and baseline increase smoothly over time. **GraphCast SEA** maintains a consistently lower error margin across all lead times.

- **u10 & v10 (10m Wind Speed components):**
    - GraphCast SEA (blue) initiates with a low RMSE (~1.1 m/s for u10) and grows monotonically due to autoregressive drift to ~2.7 m/s at 72h.
    - The Persistence baseline (red) remains distinctly higher, fluctuating between ~1.4 m/s and over 3.0 m/s.
    - *Evaluation:* The model captures boundary layer wind dynamics exceptionally well.
- **q@850hPa (Specific Humidity):**
    - Both curves progress smoothly without amplitude jitter. GraphCast SEA continuously outperforms Persistence (maintaining an RMSE of ~0.0010 to ~0.0024 kg/kg).
    - *Evaluation:* Upper-air pressure levels are less sensitive to solar radiation perturbations, allowing for a smoother optimization landscape.

Group 2: Variables Exhibiting Zigzag Patterns (Susceptible to Diurnal Aliasing)
Core Characteristic: The Persistence baseline (red) exhibits severe zigzag oscillations, creating recurring peaks (high error) and troughs (low error) tied to the diurnal cycle.

- **t2m (2m Air Temperature):**
    - The model's error increases smoothly (~0.7 K to ~1.6 K).
    - However, Persistence spikes at odd hours (>3.0 K) but plummets drastically at 24h, 48h, and 72h intervals (~1.1 K to 1.5 K), marginally outperforming the model at these exact nodes.
    - *Evaluation:* At 24-hour intervals, states are highly correlated (e.g., matching the same time of day), making the copy-paste baseline artificially strong. The model loses ground here because it lacks an explicit temporal anchor.
- **msl (Mean Sea Level Pressure):**
    - GraphCast SEA exhibits exceptional stability, with a linear RMSE progression from ~60 Pa to ~250 Pa.
    - Persistence displays a clear diurnal zigzag pattern, but fails to match the model's accuracy. GraphCast SEA retains dominance across all lead times.
- **z@500hPa (Geopotential at 500 hPa):**
    - Persistence oscillates with massive amplitude, with troughs aligning precisely at 12h intervals.
    - The model curve remains smooth. At the 24h and 48h baseline troughs, Persistence approaches or slightly edges out the model, but GraphCast dominates at all other nodes.

6.2.2 Anomaly Correlation Coefficient



ACC is the standard meteorological measure of pattern skill: the spatial correlation between forecast and observed anomalies relative to climatology:

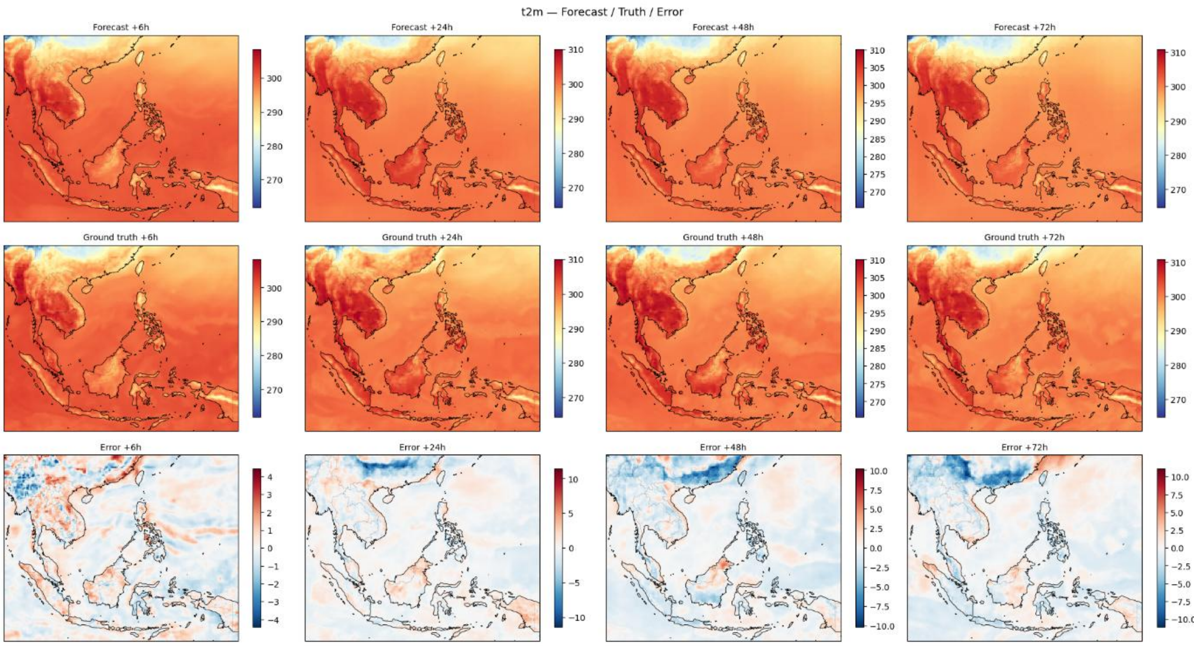
$$\mathrm{ACC} = \frac{\sum_i w_i\,(\widehat{X}_i - \bar{X}_i)(X_i - \bar{X}_i)}{\sqrt{\sum_i w_i(\widehat{X}_i - \bar{X}_i)^2}\,\sqrt{\sum_i w_i(X_i - \bar{X}_i)^2}},$$

Variable Performance Breakdown

- **t2m (2m Temperature — Blue Line): Excellent**
  - Exhibits the slowest and smoothest degradation. While it starts slightly below the pressure variables (~0.95), it remains robustly above 0.8 even at the 72h mark. The model demonstrates a highly stable capability to maintain surface temperature spatial patterns over time.
- **z@500 (500hPa Geopotential — Red Line) & msl (Mean Sea Level Pressure — Orange Line): Strong**
  - Both start with the highest initial correlation (~0.98 at 6h), indicating the model captures initial pressure dynamics extremely well. They degrade faster than t2m but safely remain above the 0.6 threshold at 72h (landing between 0.65 and 0.68).
- **u10 (10m Zonal Wind — Green Line): Acceptable**
  - Shows a steady, linear decline. It approaches and grazes the 0.6 skill threshold right around the 66h–72h mark, representing the maximum effective forecasting window for surface wind fields in the current setup.
- **q@850 (850hPa Specific Humidity — Purple Line): Weakest**
  - Starts with the lowest initial correlation (~0.88) and suffers the steepest decline. It formally drops below the 0.6 threshold between 45h and 48h,

ending near 0.5 at 72h. By day two, the predicted spatial structure of humidity is largely out of sync with reality.

6.2.3 t2m - Forecast /Truth / Error



1. Macro-Pattern Matching & Ocean Stability

- **Strong Spatial Capture:** The model successfully reproduces the macro-scale thermal distribution across Southeast Asia. It accurately pinpoints the high-temperature regional hotspots (e.g., plains of Thailand and Cambodia) and the cooler highland areas.
- **Oceanic Accuracy:** On the Error row, the oceanic regions (South China Sea, Gulf of Thailand, Andaman Sea) remain remarkably pale/white from +6h all the way to +72h. The model perfectly captures the high thermal inertia and stability of the sea surface.

2. Autoregressive Drift & The Smoothing Effect

- **Scale Jump:** The +6h forecast is highly accurate, evidenced by a tightly bounded error scale of [-4, 4]. However, by the +24h mark, error accumulation forces the colorbar to expand drastically to [-10, 10].
- **The Smoothing Artifact:** By +48h and +72h, the Forecast maps appear visibly "smoother" and more blurred compared to the granular, localized noise present in the Ground Truth. This is a common neural network phenomenon: as the model rolls out over multiple steps, it tends to output "safe" regional averages and gradually loses the ability to predict high-frequency, localized anomalies.

3. Spatial Error Distribution

- **Mountainous Cold Bias:** The most prominent errors at +48h and +72h are the deep blue patches (severe negative errors) concentrated in the high-altitude, complex terrains of the northwest (Northern Vietnam, Myanmar, and Southern China). The model is predicting temperatures significantly colder than reality. This is likely due to the coarse resolution of the topographic input (geopotential surface), which fails to capture the complex radiation absorption and retention dynamics of deep valleys and mountain slopes.

- **Landmass Instability (Diurnal Issue):** In stark contrast to the stable oceans, the landmasses on the Error row display patchy, alternating red (too hot) and blue (too cold) anomalies. Because land heats and cools rapidly throughout the day, this spatial noise visually confirms the model's struggle with the diurnal cycle; without explicit temporal features (time of day), it fails to accurately estimate the amplitude of daily solar radiation.

6.2.4 Evaluation of Metrics (RMSE) for Models and Baselines at +24h and +72h Lead Times

var	lead	GNN	U-Net	Clima	Persist
d2m	+24h	1.051	0.911	2.692	1.329
d2m	+72h	1.911	1.701	2.778	1.969
msl	+24h	121.824	108.124	307.323	150.333
msl	+72h	249.413	229.594	315.927	277.373
q@850	+24h	0.002	0.001	0.002	0.002
q@850	+72h	0.002	0.002	0.002	0.003
q@500	+24h	0.001	0.001	0.001	0.001
q@500	+72h	0.001	0.001	0.001	0.001
t2m	+24h	1.011	0.927	2.726	1.152
t2m	+72h	1.615	1.516	2.757	1.574
tp	+24h	0.001	0.001	0.001	0.001
tp	+72h	0.001	0.001	0.001	0.001
u@850	+24h	2.675	2.218	4.850	3.222
u@850	+72h	4.213	4.123	4.938	4.650
u@500	+24h	3.497	2.912	6.027	4.171
u@500	+72h	5.150	5.190	5.878	6.109
u10	+24h	1.795	1.438	3.121	2.111
u10	+72h	2.718	2.544	3.189	2.950
v@850	+24h	2.362	1.979	3.293	3.057
v@850	+72h	3.418	3.331	3.403	4.192
v@500	+24h	3.267	2.808	3.904	4.228
v@500	+72h	4.451	4.248	3.991	5.393
v10	+24h	1.832	1.488	2.981	2.398
v10	+72h	2.892	2.588	3.022	3.255
z@850	+24h	87.473	78.894	178.314	98.939
z@850	+72h	169.830	161.400	186.724	188.542
z@500	+24h	102.055	88.449	237.628	110.615
z@500	+72h	184.747	186.817	238.868	198.952

- **Strong Spatial Skill (ACC):** GraphCast SEA consistently outperforms the Persistence baseline in the Anomaly Correlation Coefficient (ACC) across all variables and lead times. It successfully maintains spatial accuracy above the critical 0.6 threshold at +72h for most variables, except for q@850hPa.

- **Short-Term Dominance:** At the initial +6h lead time, GraphCast decisively beats Persistence in both RMSE and ACC across the board.
- **The Challenge:** The table also quietly reveals that the operational model forecasts more than the 14 headline variables: d2m, tp, and the full set of u/v/z/q at both 850 and 500hPa appear in Figure 7. More importantly, the U-Net's short-range edge is an actionable signal. It suggests the GNN is currently under-resolved at small scales. The multi-scale mesh and wider hidden dimension proposed in Section 7 directly target that weakness, aiming to combine the CNN's local sharpness with the GNN's long-range, geometry-aware reach.

7. Future Plan

7.1 Data Expansion

- Increase ERA5 data to 10 years (2015–2024) — the statistical minimum required for meteorological ML models.
- Increase pressure levels from 2 to 8 levels (50–925 hPa).
- Add variables: Sea Surface Temperature (SST), Total Column Water Vapour (TCWV), and temperature at various pressure levels.
- Use actual DEM (Digital Elevation Model) from GEBCO/ETOPO datasets.
- Apply float16 format and Zarr streaming to meet Kaggle's 20 GB limit.

7.2 Model Architecture Upgrades

- Multi-scale icosahedral grid (3 scales: 8°, 4°, 2°) with cross-scale edges — significantly increasing the receptive field without adding many layers.
- Vectorize the processor using PyTorch Geometric's MessagePassing — speeding up training corresponding to batch size.
- Mixed precision training (torch.cuda.amp) — providing a 1.5–2x speedup essentially for free.
- Increase hidden dimensions to 256 or 384 and the number of layers to 8–12, incorporating skip connections.
- Target total parameters: 10–30 million (approximating the original GraphCast's 36.7 million).

7.3 Scientific Research

- **Ablation studies** — comparison tables evaluating the contribution of each component:
    - With/without the NASA satellite branch
    - With/without the multi-scale grid
    - With/without temporal features
    - With/without actual DEM
    - With/without log1p(tp) transformation
    - With/without multi-step fine-tuning
- **South China Sea typhoon case studies:** Analyze historical typhoons (Yagi

2024, Noru 2022, Rai 2021), comparing their tracks and intensities with forecasts from JMA, HKO, and IFS HRES.
- **Baseline comparisons:** Persistence, climatology, IFS HRES from ECMWF open data, and a U-Net baseline (a CNN in the deep learning family).
- **Advanced metrics:** ACC (Anomaly Correlation Coefficient — a standard meteorological metric), spectral analysis (to check for model blurring/smoothing), and regional RMSE (land vs. sea, mountainous vs. plain regions).

7.4 Deployment

- Export the model to ONNX format.
- FastAPI backend.
- Streamlit dashboard with Mapbox integration.
- Daily inference using GFS analysis as input.

8 Conclusion

GraphCast SEA demonstrates that a compact (~0.54M-parameter) Encoder → Processor → Decoder GNN, trained on dual ERA5 + satellite streams with a latitude-weighted residual objective, can deliver skillful 72-hour regional forecasts over Southeast Asia: ACC stays above the 0.6 skill threshold for most variables, and the model beats persistence and climatology at essentially all non-diurnal nodes. The error analysis localizes three well-understood limitations of autoregressive drift, MSE-induced spectral smoothing and diurnal aliasing. each with a concrete remedy in Section 7. The honest comparison against a U-Net shows the current 17 GNN is under-resolved at small scales in the short range but increasingly competitive as the forecast becomes dynamics-dominated, motivating the multi-scale, higher-capacity architecture as the natural next step toward an operational system.